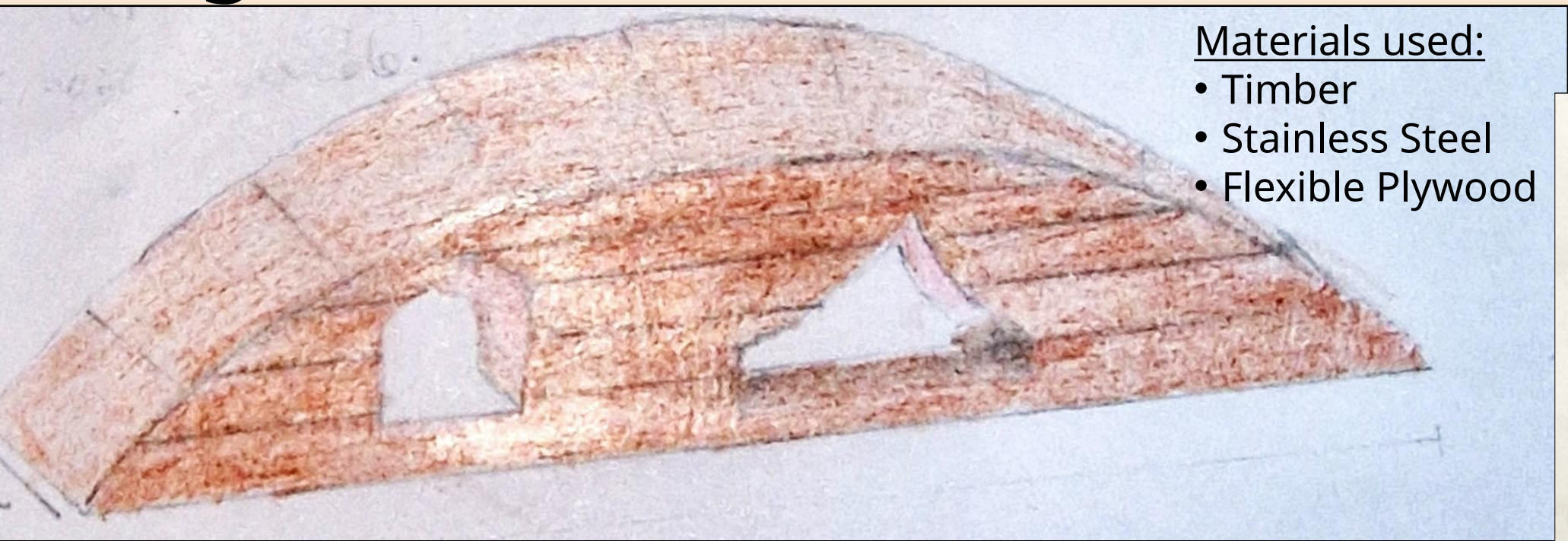


Vertical Barrier Individual Designs:

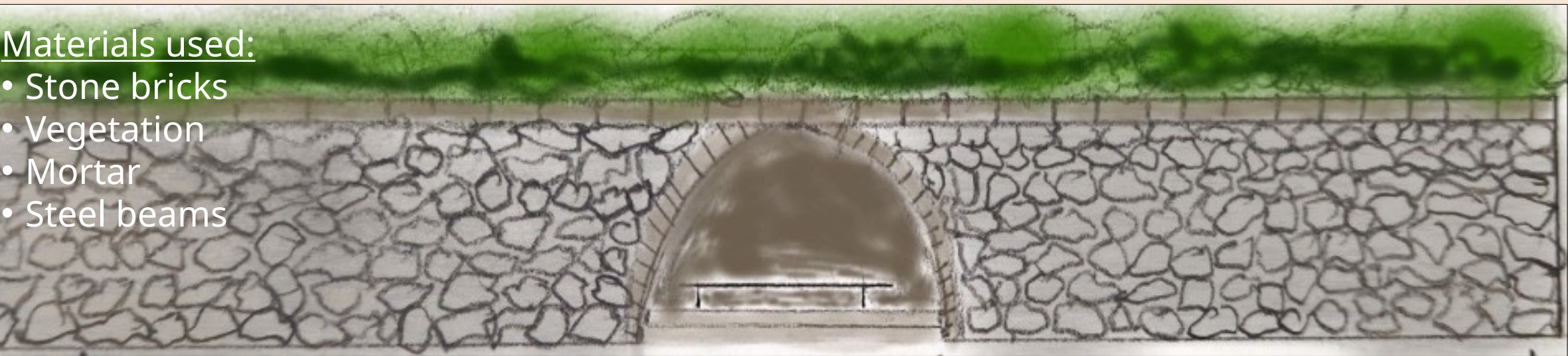


- Materials used:
- Timber
 - Stainless Steel
 - Flexible Plywood

- Advantages:
- Allows natural light and air to flow through
 - Faster and easier to construct
 - Lower materials and labour cost
 - Lightweight and a flexible design
 - 2 openings- 2 seats- makes the wall have a functional feature

- Disadvantages:
- Timber is more vulnerable compared to stone/brick (more maintenance required) especially in harsh weather conditions
 - Less structurally strong.
 - Openings allows litter to pass through

Design 2:



- Materials used:
- Stone bricks
 - Vegetation
 - Mortar
 - Steel beams

- Advantages:
- Seated area on both sides of wall allows community connection and gives function
 - Fairly quiet design doesn't overshadow or clash with historic nature of site
 - Durable design that would require little upkeep

- Disadvantages:
- Mortar takes up to 28 days to fully cure and reach maximum strength which could complicate construction
 - Costly material choices

Design 3:

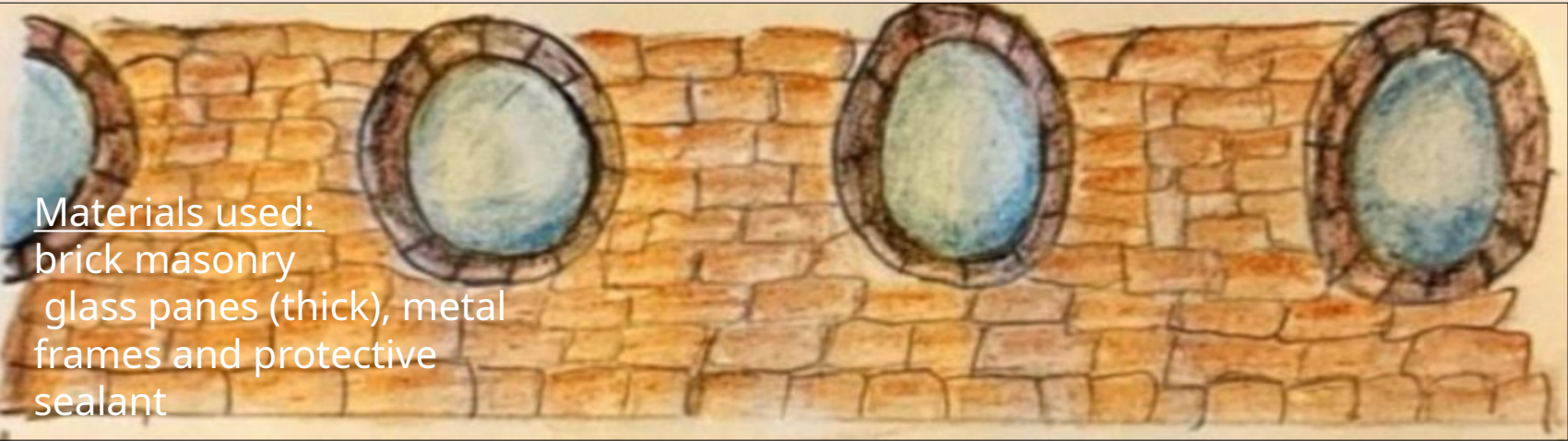


- Materials Used:
- Concrete Shipping Container
 - Steel
 - Galvanised Steel
 - Recycled Timber

- Advantages
- Concrete and steel offer strength & durability
 - Rebar and steel offers less maintenance
 - Recycled Timber and recycled shipping containers support reuse and waste reduction

- Disadvantages
- Curved structure makes it complex to build – skilled labour required.
 - Requires artistic input – more skilled labour required
 - No sustainable impact

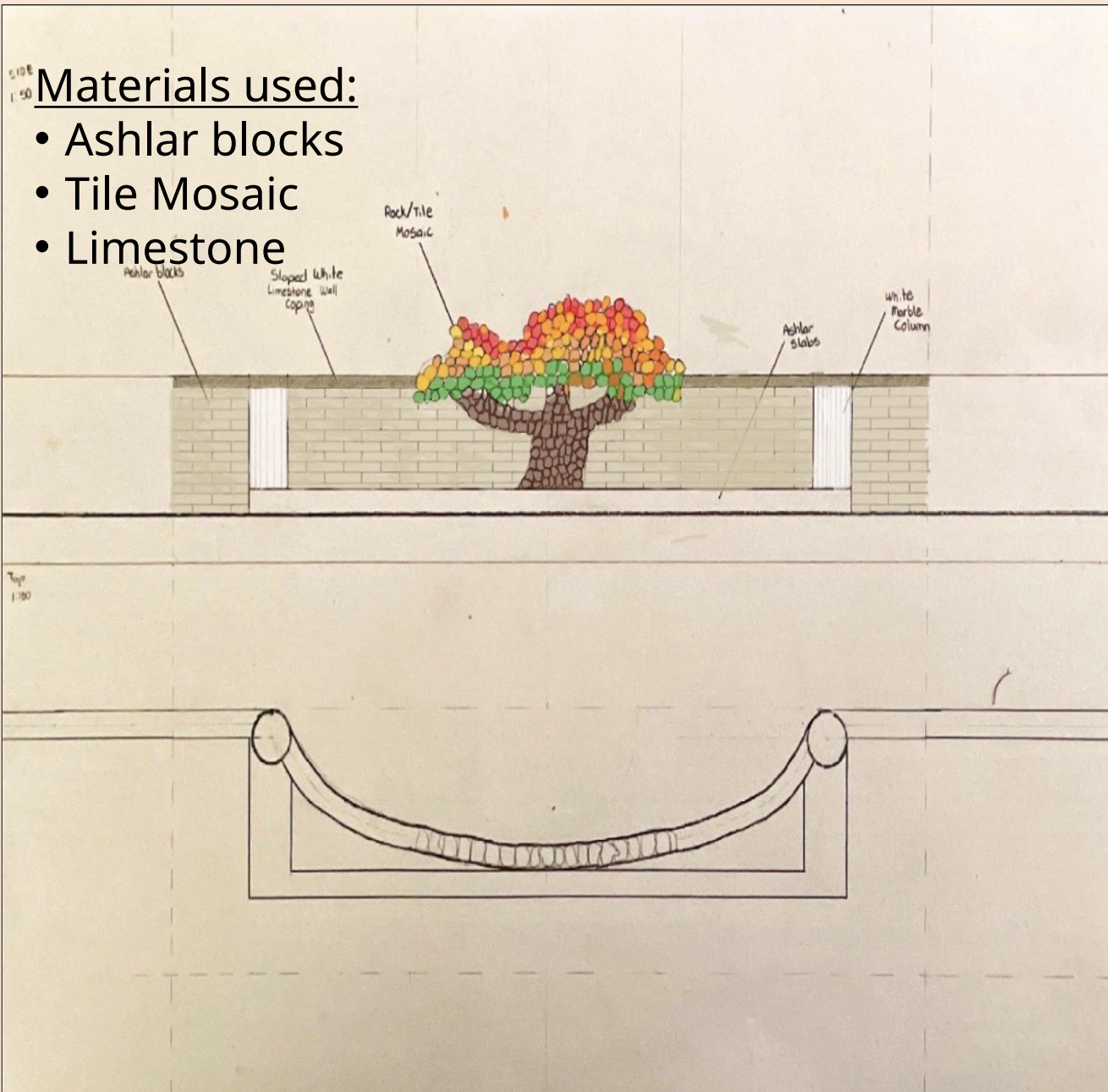
Design 4:



- Materials used:
- brick masonry
 - glass panes (thick), metal frames and protective sealant

- Advantages:
- High compressive strength
 - Good stability when properly founded
 - Resistant to fire and weathering
 - Aesthetically pleasing and links to local history and aesthetic
 - Uses locally available stone
 - Cheaper materials
- Disadvantages:
- Cannot resist large bending moments
 - Labour- intensive
 - Mortar joints deteriorate over time

Design 5:

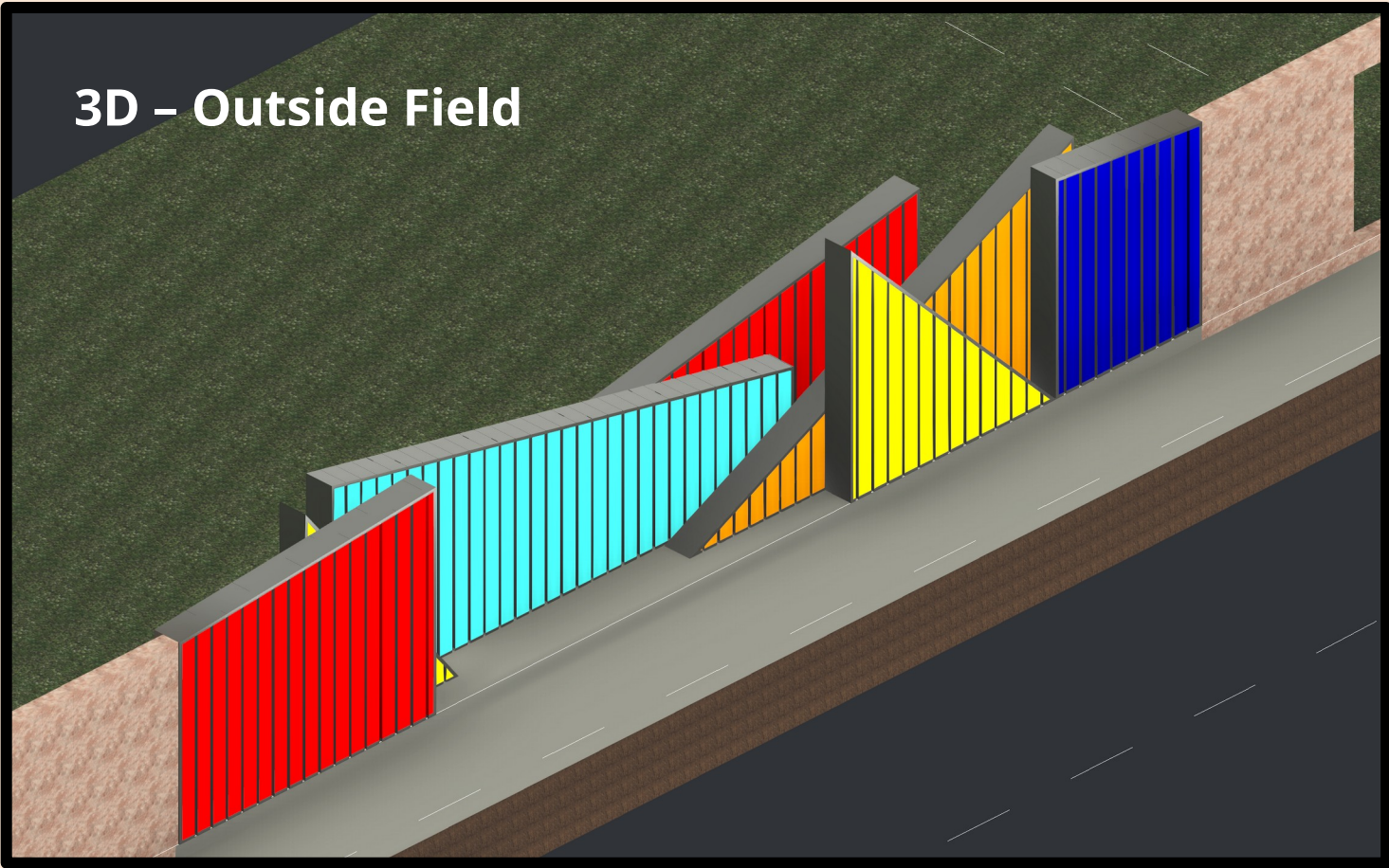


- Materials used:
- Ashlar blocks
 - Tile Mosaic
 - Limestone

- Advantages
- Low maintenance
 - Aesthetic cohesion
 - Similar materials to the site
 - Good durability

- Disadvantages
- Skilled Labour needed to cut shipping containers into specific shapes
 - Steel coating damages the site work (environment)
 - May need to use additional machinery to carry heavy loads.

Individual Design Assessment Matrix							
Individual Designs	Cost	Durability	Maintenance	Aesthetic Quality	Constructability	Sustainability	Total
1	4	2	5	4	4	4	23
2	1	5	5	4	2/3	3/4	20-22
3	5	4	5	5	4	4	27
4	4	2	3	5	4	3	21
5	2	5	4	5	1/2	3	19-20



3D – Outside Field

Site Analysis/Description

The wall is situated on the northern barrier of St George's field , adjacent to Cemetery road, to the left of the cemetery lodge entrance to the field. The site is on an area of constant elevation, with a good, level base for construction. Opposite from the site lies the centre for integrated energy research and the institute of medical and biological engineering, both owned by the University. Large construction operation may limit access to these buildings during the time it takes to erect the wall.

Original Concept

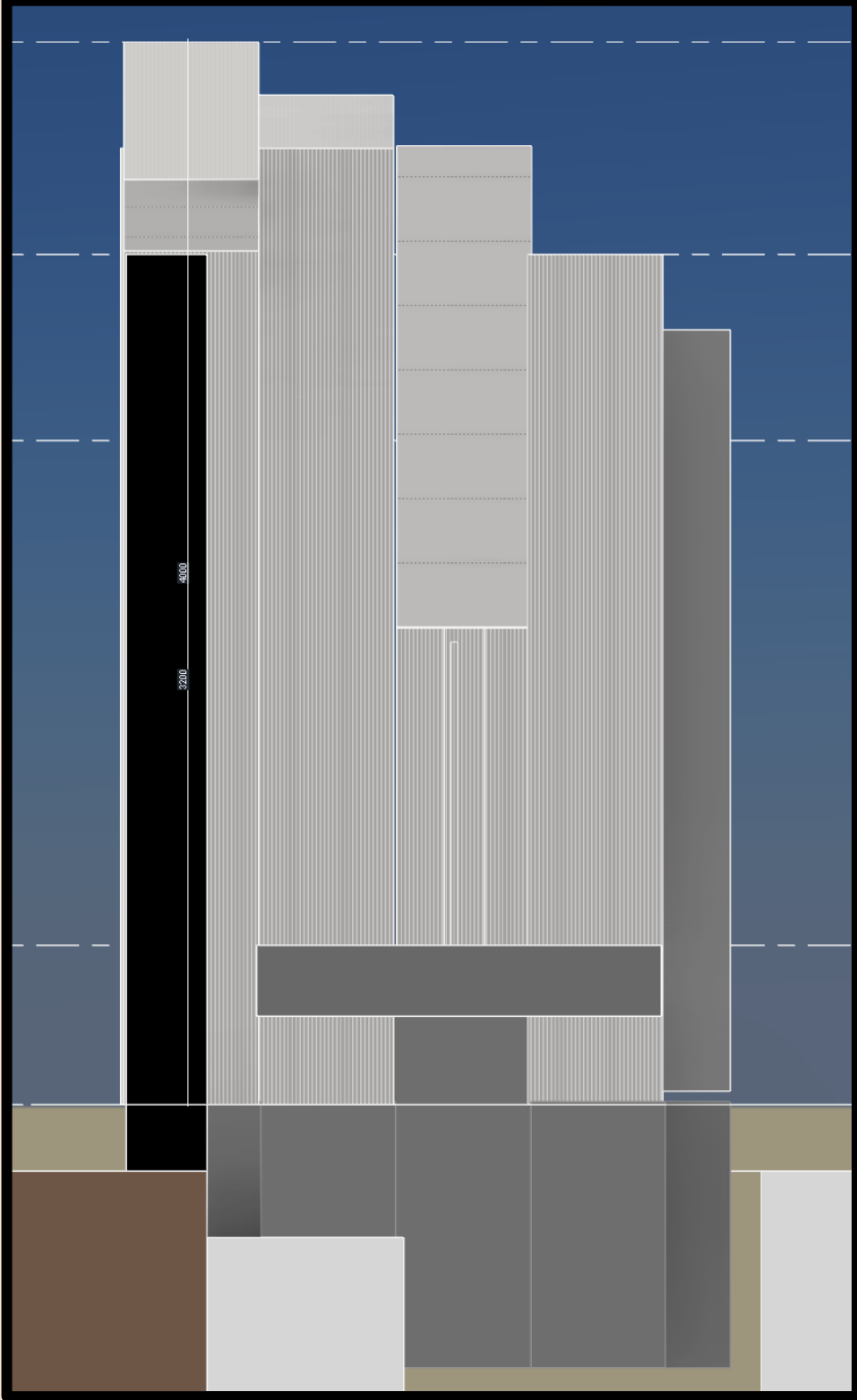
- Polygonal arrangement of custom-sized, geometric shipping-container modules.
- Angled sections designed to function as integrated planting beds.
- Bright colours and sharp angles used to create a visual break from the existing wall.
- Incorporates recycled materials and planting beds to add sustainable value.
- Aims to connect with the local community while remaining cost-effective and environmentally responsible.

Improvements in Final Scheme

- Addition of a steel frame structure to improve overall structural integrity.
- Relocation of planting beds for better practicality and visual balance.
- Introduction of benches to increase

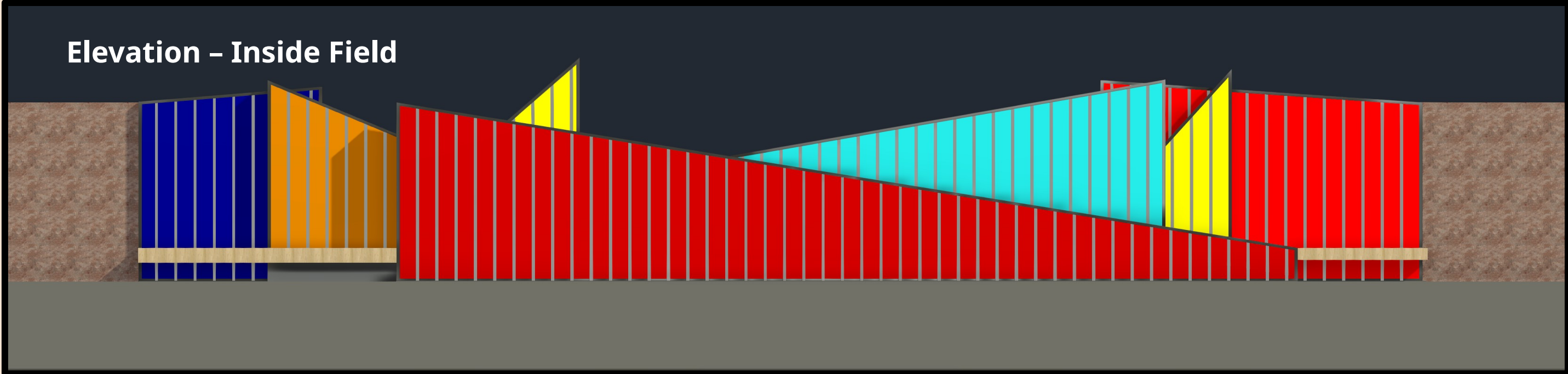
space at St.

Cross-Section View



Construction Method

- 1.Excavate 3 meters of dirt within the 5x20 meter area
- 2.Install Framework for foundation and mark parameters for each container module
- 3.Steel Frame Columns are put in place
- 4.Foundation Poured
- 5.Shipping container sheet cladding welded onto steel frame
- 6.Joints are connected to wood seat
- 7.Wood seat joint welded to steel frames
8. Custom shaped planting pots are constructed and placed on ground



Elevation – Inside Field

Material Selection and Cost Analysis

Cladding

- **Project Requirement:** ~220 m² of shipping container metal.
- **Primary Choice: Two 12m Containers** (Best cost-material ratio)
- **Cost Efficiency:** 12m units have a higher price ceiling (starting at £1100) compared to the multiple 6m units required to meet the same area (starting at £860 each).
- **Material Yield:** Two 12m containers provide 272.4 m² of metal. Resulting in about 52.4 m² in surplus.
- **Reasoning:** The surplus may be useful for covering welding seams, structural reinforcements, and accommodating for potential cutting errors without incurring additional costs.
- **Approximate Price: £5200** includes material, fabrication, and installation

Frame

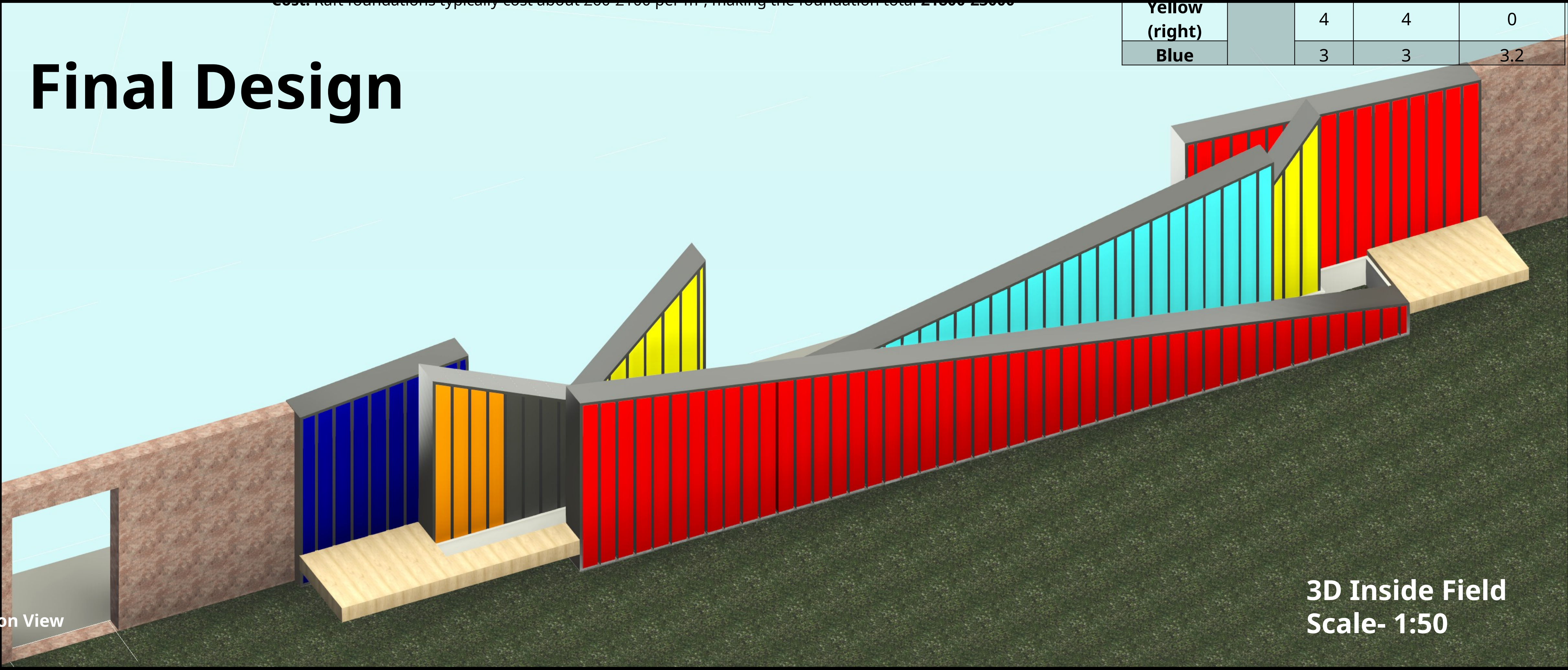
- **Project Requirement:** ~ 1 tonne of steel
- **Choice of Steel:** Weathering Steel/ Corten Steel to prevent rusting.
- **Approximate Price: ~£9000** (this includes raw materials, fabrication, finishing, and installation)

Foundation

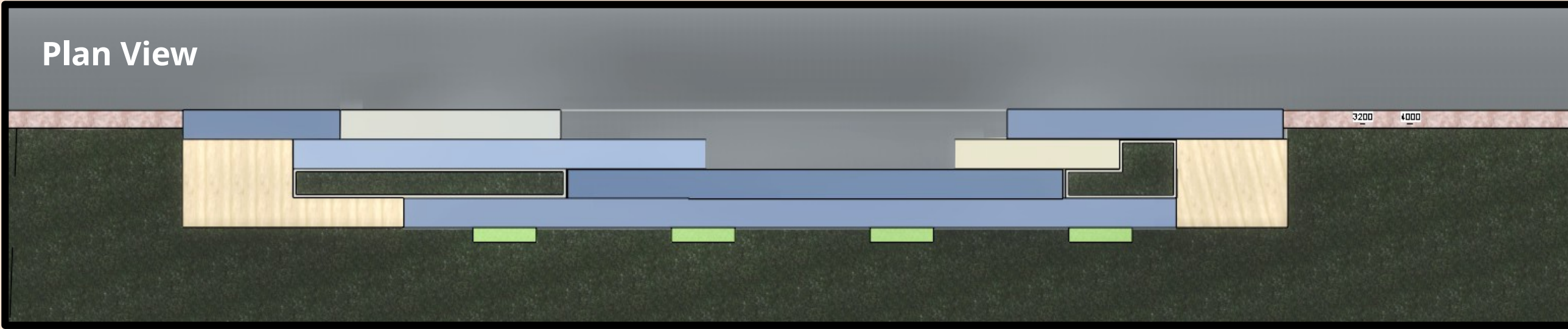
- **Total Area Footprint:** 40 m²
- **Suggested Foundation:** Raft that accommodates roughly ~30 m² (cut so foundation isn't visible from inside the park)
- ~~Cost:~~ Raft foundations typically cost about £60-£100 per m², making the foundation total ~~£1800-£3000~~

Project Cost Estimate: around **£14,000**

Dimensions Table (Relative to the Sidewalk Elevation)				
Main Member	Depth (m)	Base (m)	Max Height (m)	Min Height (m)
Foundation	0.5-2	20	0.4	0
Red (left)	0.5	5	3.6	3.2
Yellow (left)		3	3.8	0
Light Blue		9	3.6	1.8
Red (center)		14	3.2	0.6
Orange		7.5	3.6	0
Yellow (right)		4	4	0
Blue		3	3	3.2



3D Inside Field Scale- 1:50



Plan View

Elevation – Outside Field

